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Scientific and Production Association
"KASKAD" Ltd.

Approved
Scientific and Production Association "KASKAD" Ltd.
Director General

ОБЩЕСТВО С ОГРАНИЧЕННОЙ ОТВЕТСТВЕННОСТЬЮ НАУЧНО-ПРОИЗВОДСТВЕННОЕ ОБЪЕДИНЕНИЕ «КАСКАД»

_____________V.B. Voronov

FIXED AEROSOL FIRE-EXTINGUISHING SYSTEM

(AF SYSTEM)

PROGRAM AND METHOD OF ACCEPTANCE TEST

PM AT AFS KASKAD

ГЛАВНОЕ УПРАВЛЕНИЕ
HEAD OFFICE
PC RS
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Contents

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TESTING PROGRAM

The testing program and method is assigned for conducting commissioning tests of a fixed aerosol fire-extinguishing ship system (AFS) manufactured by Scientific and Production Association "KASKAD" Ltd.

Acceptance tests of AF system should be conducted by the manufacturer or an organization certified by SPA “KASKAD” Ltd.

Testing object

Ship aerosol fire-extinguishing system (AFS) extinguishes fire in ship premises in accordance with approved specifications.

Scope of testing

- Visual inspection.
- Testing at “System monitoring” mode.
- AF system actuation test (using simulators).
- AF system efficiency test at shortcut in the starting circuit.

METHOD OF TEST

1 TEST PREPARATION

AFS acceptance test should be conducted at vessel anchorage.

To conduct an acceptance test the following equipment is required:

- a multimeter;
- a megaohmmeter;
- simulators of starting assemblies manufactured by SPA KASKAD Ltd. in the number equal to the total number of aerosol fire-extinguishing system generators.

In case of absence of the simulators the following equipment is required:

- resistors with resistance equal to the one of the starting assembly of AF generators (the data are given in generator specification);
- incandescent lamps with supply voltage of 24 V and design current of 1 A.

Before conducting the test check:

- the insulation resistance of all AFS circuits;
- the conformity of supplied voltage to the required voltage (24±8 VDC).

Make sure that generators are disconnected from the starting circuits.

“POWER” and “START” tumblers on a CSP AFS front panel are to be in “OFF” position.

Simulators of starting assemblies manufactured by SPA KASKAD Ltd. are to be connected to the starting circuits of each generator. The simulators are to be switched to the operative condition mode. In case of absence of simulators connect the resistors with resistance equal to the one of the starting assembly.

Supply voltage of 24 VDC to the control and signaling panel (CSP) and to the board of intermediate relays (BIR). Check the indicators “POWER 24VDC CSP” and “POWER 24VDC BIR” (if available) are lit up on the CSP front panel.

2 TEST OPERATION

2.1 Visual inspection

Conduct a visual inspection of:

- the conformity of AF system mounting to the assembly drawing;
- AF system equipment security of attachment;
- the conformity of mounting and cable marking to the electrical connection layout;
- the possibility of a complete opening and fixation of CSP, BIR and connection boxes covers;
- the installation of aerosol fire-extinguishing generators in protected premises in accordance with the following requirements:

a) aerosol sprays, formed when the aerosol fire-extinguishing generator operates, should not influence escape routes, shipboard equipment, cable paths, pipelines, oil fuel tanks and lubricating oil tanks;

б) minimal distance, measured along the axis of aerosol sprays diffusion, from an aerosol fire-extinguishing generator to the nearest surface (of the equipment or casing) or other places mentioned in paragraph *A* should not be less than 2.0 m for SOT-1M generator and less than 0.5m for SOT-2M, SOT-2M-KV and AGS-8/1M generators.

2.2 Testing at "System monitoring" mode

Check the indicators "POWER 24VDC CSP" and "POWER 24VDC BIR" on the CSP front panel are lit up. It confirms that power supply of 24VDC is available in the input terminals of the equipment. "POWER" and "START" tumblers are to be in "OFF" position.

Switch the "POWER" button to the "ON" position. At the moment of power supply a built-in CSP buzzer is to sound. After that on LCD the following report appears:

SPA «KASKAD»
AFS CSP
$N/G_1-...-...-G_N$ Vxx

where:

- $N/G_1-...-...-G_N$ - CSP configuration determining the quantity **fire-extinguishing lines*** (N) and the quantity of generators in each line ($N/G_1-...-...-G_N$);
- Vxx – software version.

****

***Fire-extinguishing line** (NN) means one or several adjacent or integrated premises at which fire-extinguishing is carrying out simultaneously. When AF system is actuated all premises being a part of one line will be filled by fire-extinguishing aerosol.*

CSP checks the continuity of starting assemblies in automatic mode with successive indication of all faulty circuits. In case of absence of faults on LCD the following report appears:

```
           AF system
          is operable
/
```

In the bottom left corner of LCD the rotating symbol «/» informs about normal operation of CSP.

If a faulty circuit is found, the built-in CSP buzzer starts to sound and on LCD the following report appears:

```
L NN  (line name)
     High R start. circuit
G
                              A
```

where:

- NN - number of a faulty line;
- G - number of a faulty circuit in Line NN;
- A - symbol of automatic system monitoring mode.

****

Example

If the circuits №2 and №4 in line 01 break, the following report appears:

```
L01  (line name)
     High R start. circuit
02 04
                              A
```

where:

- 01 – number of a faulty line;
- 02 04 – numbers of faulty circuits in line 01;
- A - symbol of automatic system monitoring mode.

****

When any of ◂ , ▸, ▴, ▾ button is pressed, CSP switches to manual system monitoring mode, and symbol "A" changes to symbol "M" (manual mode). Subsequent pressing of ◂ , ▸, ▴, ▾ allows to get access to the information about all faults. CSP can be turned to automatic system monitoring mode by pressing ↵.

To switch off/on the built-in buzzer press 🔇 on the front panel of the CSP, after which "◀" symbol appears and disappears on LCD.

While switching each of the simulators to required conditions (normal operation - break – short circuit – normal operation) make sure that faulty starting circuits and types of failures are indicated correctly (short circuit, break).

In case of absence of the generators in order to check starting circuits in different conditions:

- connect the resistor with the resistance equal to the one of starting assembly of a generator to imitate normal condition of the circuit;
- disconnect the resistor to imitate the breaking of circuit;
- short-circuit current-carrying wires of the starting circuit of the generator to imitate short circuit.

To test the condition of the control circuits of warning alarm activation, and ventilation and other devices deactivation (if available by AFS project):

- activate ventilation and other devices that are deactivated in protected premises at AFS actuation;
- press down and hold during the test.

At this moment the built-in CSP buzzer starts to sound and the following report appears on LCD:

```
ATTENTION! TEST
equipment switch off
and alarm switch on
TEST in  XX sec
```

where:

XX – countdown time in seconds until control voltage is supplied to test the operation of the circuits and the equipment mentioned above.

On expiry of programmed time control voltage is supplied from CSP and the following message if displayed:

```
TEST
```

The result of the test is considered to be positive if a warning alarm of the system actuation turns on and ventilation and other devices (if available) are deactivated in the protected premises.

Once the test is completed, switch the “POWER” button to the “OFF” position.

2.3 AFS actuation test (using simulators)

AFS test is carried out by the system actuation using simulators.

Before AFS test is conducted, simulators of starting assemblies manufactured by SPA KASKAD Ltd. are to be connected to starting circuits of each of the generators. Turn the imitators to operative condition. In case of absence of resistors use incandescent lamps (see Section 1).

2.3.1 AFS actuation test with time delay

Check that the indicators “POWER 24VDC CSP” and “POWER 24VDC BIR” on the CSP front panel are lit up. It confirms that power supply of 24VDC is available. “POWER” and “START” tumblers are to be in “OFF” position.

Switch the “POWER” button to the “ON” position. In case incandescent lamps are used instead of the simulators of starting assemblies, ignore CSP messages about faulty starting circuits. Open the lock and turn the “START” tumbler of the **first** line to the “ON” position. If a supplementary blocking unit for the activation of a chosen line is provided, insert a key into the blocking unit and to turn it clockwise up to the stop to the “I” position. The built-in buzzer starts to sound at this time and the following report appears on LCD:

```
L 01    (line name)
       AF system start
       LEAVE THE SPACE
          XX sec
```

where:

01 – number of an actuated line (first line);

XX - countdown time in seconds before starting impulses are supplied.

At the same time the warning alarm is to be activated and ventilation and other devices (if available) are to be deactivated in the protected premises of the first line.

At the moment of starting impulses are supplied the built-in CSP buzzer switches to an even sound and the following report appears on LCD:

```
L 01    (line name)
        AEROSOL START!!!

          GA ... ... GD
```

where:

01 – number of an actuated line;

G_A ... ... G_D – numbers of generators actuated at the moment.

In the protected premises of the first line check the simulators light up alternately by groups each including 4 pieces.

After starting impulses are supplied to all the starting circuits of the first line the following report appears on LCD:

```
L 01    (line name)
          AEROSOL is
         in the space
           XX  min
```

where:

01 – number of an actuated line;

XX – time delay in minutes (countdown from 50 to 0 minutes)

Countdown is kept up from 50 to 0 minutes, afterwards the following report appears on LCD:

```
L  01          (line name)
          SWITCH OFF
        starting tumbler
```

It is not required to check this report to appear.

2.3.2. AF System emergency mode

AF system emergency mode is enabled after AF system start check with starting time delay without any change in the condition of CSP boarding panel.

To check the emergency mode of AF system press the «EMERGENCY MODE» buttons of the required line successively one after another holding each pressed for 1-2 seconds. In protected premises of the first line check all simulators of starting assemblies (incandescent lamps) light up successively.

Each button of emergency mode supplies starter current to the relative group including 4 or less simulators. The period of time it takes simulators to light up depends on how long «EMERGENCY MODE» button is in pushed position.

ATTENTION

To avoid overloading of AFS power circuits it is strongly prohibited to:

- Simultaneously press the “EMERGENCY START” buttons for several times;
- Hold “EMERGENCY START” buttons when the following report appears on LCD:

```
N NN   (line name)
        AEROSOL START!!!

             GA ... ... GD
```

While the actuation of AF system test it is necessary to check:

- the indicator on CSP front panel;
- the built-in CSP buzzer functioning;
- activation of the warning alarm of AFS actuation in protected premises;
- deactivation of ventilation and other devices in protected premises;
- programmed time delay before simulators light up;
- that simulators within the checked line light up successively during system start with starting time delay;
- that simulators within the checked line light up successively during system start with emergency start;
- efficiency of the supplementary blocking unit of the system actuation (if available).

It is required to test AFS actuation of other lines in a similar manner.

After completion of the check turn the “START” and “POWER” tumblers to the “OFF” position. If a supplementary blocking unit for the activation of a chosen line is provided, turn the inserted key counterclockwise up to the stop at the “O” position and take out the key.

2.4. Testing of AFS efficiency at shortcut in the starting circuit

Fix a bridge on terminal blocks of a generator starting circuit. Switch the “POWER” button to the “ON” position. Set the “START” tumbler of a chosen line with shunted generator to the “ON” position. If a supplementary blocking unit for the activation of a chosen line is provided, insert the key into the blocking unit and turn it clockwise up to the stop to the “I” position.

When a programmed time delay has passed check that all of the simulators of the chosen line except for the bridged simulator light up successively.

Switch the “START” and “POWER” buttons to the “OFF” position. Take off the bridge from the generator starting circuit and actuate the same line for the second time.

Test results are considered to be positive if:

- at the first actuation all the simulators except for the bridged one light up successively;
- at the second actuation of the chosen line all the simulators including the previously bridged one light up successively in a protected space.

After completion of AFS test turn the “POWER” and the “START” tumblers on the CSP front panel to the “OFF” position. If a supplementary blocking unit for the activation of a chosen line is provided, insert the key into the blocking unit and turn it clockwise up to the stop to the “O” position and take the key out.

“START” tumbler is to be sealed using a wire which can be breaked off by hand. Seal number is to be registered in the AFS commissioning test record.

Disconnect the starting assembly simulators from the starting circuits. Connect the fire-extinguishing generators to the starting circuits.

Make sure of the generators starting circuits continuity by switching the CSP to the “System monitoring” mode. Turn the “POWER” tumbler to the “OFF” position.

AFS acceptance test results are to be registered in the Acceptance tests report (Annex 1). A copy of the record is to be sent to Scientific and Production Association "KASKAD" Ltd.:

tel./ fax +7 (812) 335-66-88, 335-66-87

mail@kaskad.net.ru

Annex 1.

ACCEPTANCE TESTS REPORT
fixed marine aerosol fire-extinguishing system (AFS)

"________" ______________________ 20______

ship name		
project	*register, class*	*register number /IMO number*

AF system specification	*model*	*serial number*
Control and signaling panel CSP		
AF system generators	*quantity*	*lot*
SOT-1M		
SOT-2M		
SOT-2M-KV		

By the results of the acceptance tests the AF system operation is

Allowed	**Allowed with limitations** (limitations stated on the form's reverse side)	**Not allowed**

(cross out unnecessary)

"START" tumblers are sealed. Register numbers of seals____________________________________

Next AF system examination is to be conducted no later than:

______________________________**20**________
(in five years from the date of the AF system acceptance tests)

Company	Surname Name	Signature

Send copy to SPA "KASKAD" Ltd.:
tel./ fax +7 (812) 335-66-88, 335-66-87, mail@kaskad.net.ru, www.kaskad.net.ru